

Data dictionary

Beverage distributor with own fleet — synthetic dataset

16 tables · Period 2022-01-01 to 2024-12-31

Catalog

Depositos

Branches/warehouses the distributor operates from.

Field	Type	Key	Description
deposito_id	int	PK	Unique warehouse identifier.
nombre	string		Warehouse name (city/branch).
localidad	string		Locality where the warehouse is located.

Modelos_Vehiculo

Technical catalog of vehicle models (factory specs). Multiple physical units in Vehiculos can share the same model.

Field	Type	Key	Description
modelo_id	int	PK	Unique model identifier.
marca	string		Manufacturer brand.
modelo	string		Model name/version.
tipo	string		Vehicle type: truck or van.
potencia_hp	int		Engine power in horsepower.
cilindrada_cc	int		Engine displacement in cubic centimeters.
transmision	string		Transmission type (e.g. manual_6).
tipo_combustible	string		Fuel type used by the model.
capacidad_tanque_l	float		Fuel tank capacity, in liters.
tara_kg	float		Empty vehicle weight, in kilograms.
capacidad_carga_kg	float		Maximum cargo capacity, in kilograms.
capacidad_carga_m3	float		Maximum cargo capacity, in cubic meters.
configuracion_eje	string		Axle configuration (e.g. 4x2, 6x2).
consumo_teorico_l_100km	float		Factory-declared fuel consumption, in liters per 100km.

Zonas

Delivery areas within each warehouse's radius.

Field	Type	Key	Description
zona_id	int	PK	Unique zone identifier.
nombre	string		Zone/neighborhood name.
deposito_id	int	FK	Warehouse this zone belongs to.

Productos

Catalog of beverages the company distributes.

Field	Type	Key	Description
producto_id	int	PK	Unique product identifier.
nombre	string		Commercial product name.
categoria	string		Category (soda, beer, water, juice, sports drink).
presentacion	string		Sale format (e.g. 354ml can, 6-pack 500ml).
peso_kg	float		Total package weight, in kilograms.
volumen_l	float		Total package volume, in liters.
precio_base	float		Reference price at the start of the period (2022-01-01), before inflation.

Insumos

Catalog of parts and supplies used in maintenance.

Field	Type	Key	Description
insumo_id	int	PK	Unique supply/part identifier.
nombre	string		Supply/part name (e.g. engine oil, tire).
unidad_medida	string		Unit of measure (liter, unit, set, kit, pair).
costo_unitario_referencia	float		Reference cost per unit, before inflation.

Commercial

Clientes

Businesses the distributor sells to (kiosks, supermarkets, etc.).

Field	Type	Key	Description
cliente_id	int	PK	Unique client identifier.
nombre	string		Client name/business name.
tipo_cliente	string		Business type (kiosk, supermarket, bar/restaurant, wholesaler, self-service store).
zona_id	int	FK	Delivery zone where the client is located.
fecha_alta	date		Date the client was added to the portfolio (may precede the start of the period).
fecha_baja	date		Date the client stopped buying (empty if still active). There are no orders after this date.

Pedidos

Header record for each order placed by a client.

Field	Type	Key	Description
pedido_id	int	PK	Unique order identifier.
cliente_id	int	FK	Client who placed the order.
fecha_pedido	date		Date the order was placed.
fecha_entrega_estimada	date		Estimated delivery date.
estado	string		Order status: delivered, cancelled or pending.

Detalle_Pedido

Product lines for each order (composite key: pedido_id + producto_id).

Field	Type	Key	Description
pedido_id	int	PK/FK	Order this line belongs to.
producto_id	int	PK/FK	Product ordered in this line.
cantidad	int		Quantity of packages ordered.
precio_unitario	float		Unit price in effect at order time (already includes accumulated inflation).

Fleet

Vehiculos

Physical fleet units (each linked to a technical model and a warehouse).

Field	Type	Key	Description
vehiculo_id	int	PK	Unique vehicle identifier.
patente	string		Vehicle license plate.
modelo_id	int	FK	Technical model of this vehicle.
deposito_id	int	FK	Warehouse this vehicle is assigned to.
km_inicial_periodo	float		Estimated odometer reading as of 2022-01-01 (or the registration date if later), based on its age.
km_actual	float		Most recent odometer reading in the system (intentionally outdated for a % of vehicles).
fecha_ultima_actualizacion_km	date		Date km_actual was last updated.
estado_actual	string		Current status: operational, in workshop, or out of service.
fecha_alta	date		Date the vehicle joined the fleet.
fecha_baja	date		Vehicle decommission date (empty if still active).

Choferes

Drivers employed by the distributor.

Field	Type	Key	Description
chofer_id	int	PK	Unique driver identifier.
nombre	string		Driver's full name.
fecha_ingreso	date		Date the driver joined the company.
categoria_licencia	string		Qualifying driver's license category.
deposito_id	int	FK	Warehouse this driver is assigned to.

Historial_Estado_Vehiculo

Historical log of each vehicle's status changes (in workshop / out of service), with a free-text reason.

Field	Type	Key	Description
historial_id	int	PK	Unique history event identifier.
vehiculo_id	int	FK	Vehicle this event corresponds to.
fecha_desde	date		Start date of the status.
fecha_hasta	date		End date of the status (empty = currently in effect).
estado	string		Recorded status: in workshop or out of service.
motivo	string		Free-text reason (may have formatting variants: uppercase, abbreviations, etc.).

Operations

Viajes

Delivery route run by a vehicle and driver on a given date and zone, grouping several deliveries.

Field	Type	Key	Description
viaje_id	int	PK	Unique trip identifier.
vehiculo_id	int	FK	Vehicle that performed the trip.
chofer_id	int	FK	Driver who drove the trip.
zona_id	int	FK	Zone covered by the trip.
fecha	date		Date the trip took place.
km_inicial	float		Vehicle odometer at the start of the trip.
km_final	float		Vehicle odometer at the end of the trip.
hora_salida	string		Departure time from the warehouse.
hora_llegada	string		Return time to the warehouse.

Viaje_Pedido

N:M bridge table between Viajes and Pedidos: which orders were delivered on each trip and in what order.

Field	Type	Key	Description
viaje_id	int	PK/FK	Trip on which the order was delivered.
pedido_id	int	PK/FK	Order delivered on this trip.
orden_parada	int		Position of this delivery within the trip's route.
hora_entrega_real	string		Actual time the delivery took place.

Costs and maintenance

Combustible

Fuel loads performed for each vehicle.

Field	Type	Key	Description
carga_id	int	PK	Unique fuel load identifier.
vehiculo_id	int	FK	Vehicle that was refueled.
fecha	date		Date of the fuel load.
litros	float		Liters loaded (may include deliberate outliers, ~3% of cases).
precio_litro	float		Diesel price per liter in effect on that date.
km_odometro	float		Vehicle odometer at the time of the load.

Mantenimiento

Preventive or corrective maintenance events performed on each vehicle.

Field	Type	Key	Description
mantenimiento_id	int	PK	Unique maintenance event identifier.
vehiculo_id	int	FK	Vehicle that received the maintenance.
fecha	date		Date of the event.
tipo	string		preventive or corrective.
subtipo	string		Detail of the preventive service (oil/filter change, brakes, tires, timing belt); empty for corrective events.
km_odometro	float		Vehicle odometer at the time of the event.
taller	string		Workshop where the work was performed.
costo_mano_obra	float		Labor cost of the event (already includes accumulated inflation).

Detalle_Mantenimiento

Supplies/parts used in each maintenance event.

Field	Type	Key	Description
mantenimiento_id	int	FK	Maintenance event this line belongs to.
insumo_id	int	FK	Supply/part used.
cantidad	float		Quantity used, in the supply's unit of measure.
costo_total	float		Total cost of this supply in this event (already includes accumulated inflation).

Additional note

Reference table `inflacion_trimestral.csv`: not part of the relational model (no incoming or outgoing FKs). Stores, per quarter, the sampled inflation rate and the accumulated factor used to compute `precio_unitario`, `precio_litro` and `costo_mano_obra/costo_total` across the other tables. Included as supporting documentation so values can be deflated during analysis.